

Experimental and Theoretical Modeling of Polarization-Mode Dispersion in Single-Mode Fibers

A. O. Dal Forno, A. Paradisi, R. Passy, and J. P. von der Weid

Abstract—The random coupling-length basis was used in the numerical simulation of single-mode fibers (SMF's) with the coarse step method. We show that a single set of coupling lengths, angles and phases accounts for both, the Maxwellian statistics and the nonperiodical differential group delay spectral dependence in agreement with experimental observations. An SMF and a polarization mode dispersion (PMD) emulator were both measured and simulated. The comparison between the experimental results and numerical simulations shows that the random coupling-lengths mathematical model and the emulator device provide good descriptions either for the first-order PMD statistics or second-order PMD, being powerful tools for the simulation of signal distortion.

Index Terms—Optical communications, polarization mode dispersion.

I. INTRODUCTION

POLARIZATION mode dispersion (PMD) still attracts considerable attention due to its potential limitation on optical transparent high-speed long-distance lightwave systems, as well as on multichannel cable television (CATV) transmission systems. PMD in long single-mode fibers (SMF's) has a stochastic characteristic, because of random-mode coupling induced by length dependent random birefringence [1], [2]. The differential group delay (DGD) between the principal states of polarization (PSP) varies in time and wavelength, a phenomenon known as second-order PMD, which introduces additional signal distortion [3]. For high random-mode coupling fibers, the DGD's have a well-known Maxwellian statistical distribution [1], [4].

Emulators are commonly used to understand PMD effects in optical lightwave systems, as well as to test theoretical simulations or compensating devices [5], [6]. They are usually made with single pieces of high-birefringence (HiBi) fibers or with a set of many concatenated equal segments of those fibers [6], [7]. Numerical simulations with the coarse-step method also consider an SMF as a concatenation of equal length segments with a given mean birefringence and random coupling angles [8], [9]. This conception gives rise to a set of equally spaced time-of-flight values along the modeled fiber [10], a situation that certainly differs from the reality of a long SMF, even though the statistics of the PMD is well described by the model. A different

approach to design PMD emulators considered a SMF as a concatenation of many HiBi fiber segments with different lengths and coupling angles, accounting for a random time-of-flight distribution along the device [10].

Numerical simulations that account for the actual wavelength dependence of the PMD effects are of the greatest importance for the prediction of the PMD distortion and penalties in optical systems. Moreover, PMD emulator devices that correctly describe these effects are powerful tools for testing the sensitivity of optical systems and components to PMD. In fact, though these effects are statistically described by the Maxwellian distribution, the actual DGD values and its wavelength dependence induce the distortion, so that the model must predict realistic situations before the statistics is performed. In this work we investigate PMD effects that occur within the spectral window of a WDM channel, attempting to describe the instantaneous situation that a chirped optical signal will find when propagating along the fiber. We include a random coupling-lengths basis in the coarse step method to account for the random time-of-flight distribution and discuss the conditions and limitations of the solution. Finally we test the DGD spectral dependence and statistical predictions of the concatenated HiBi PMD emulator, comparing both the experimental and theoretical model results with a real fiber with the same total PMD.

II. THEORETICAL MODELING

The Jones matrix $T(\omega)$ that describes a concatenation of unequal sections of birefringent fiber can be calculated by N products of two matrices:

$$\begin{aligned}
 T(\omega) &= \prod_{n=1}^N B_n(\omega) R(\alpha_n) \\
 &= \prod_{n=1}^N \begin{bmatrix} e^{j(\sqrt{3\pi/8} b \omega \sqrt{h_n}/2 + \phi_n)} & 0 \\ 0 & e^{-j(\sqrt{3\pi/8} b \omega \sqrt{h_n}/2 + \phi_n)} \end{bmatrix} \\
 &\quad \cdot \begin{bmatrix} \cos \alpha_n & \sin \alpha_n \\ -\sin \alpha_n & \cos \alpha_n \end{bmatrix} \quad (1)
 \end{aligned}$$

where

- N is the number of segments;
- $B_n(\omega)$ represents the birefringence matrix of n th segment with h_n length;
- $R(\alpha_n)$ is the matrix of a rotator that represent the random coupling angle between the segments axes;
- b is the PMD coefficient of the fiber in ps/ $\sqrt{\text{km}}$;
- ω is the optical frequency.

Manuscript received September 7, 1999; revised November 10, 1999. This work was supported in part by the CPqD Foundation under the Contract CTO/P&D/067/98 and the Brazilian National Research Council CNPq.

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Publisher Item Identifier S 1041-1135(00)01983-2.

We consider for simplicity different lengths, keeping the same mean value of the birefringence for all the segments. The phase ϕ_n accounts for the small temperature fluctuations along the fiber, being a stochastic variable with a uniform distribution between 0 and 2π . This approach is equivalent to the random phase used in [9]. The DGD $\delta\tau(\omega)$ for a single wavelength is calculated from the matrix $T(\omega)$ with the method described in [11] by calculating the eigenvalues of the matrix $\partial_\omega T(\omega)T^{-1}(\omega)$.

Simulations were performed for concatenations of $N = 20, 50, 100$, and 200 sections. For a given total PMD $\langle\Delta\tau\rangle$ and fiber length L the size of each segment was randomly generated from a Gaussian distribution around the mean length L/N with standard deviations Δh varying from 0–30% of the mean length. In order to evaluate the amount of scattering in the section lengths needed to eliminate the periodicity in the DGD spectrum we searched for peaks in its Fourier transform. Fig. 1 presents the case of a fiber with 27 ps PMD and 50 sections for different standard deviations Δh . For equal sections we observe that the spectrum presents peaks at multiples of 4 ps, which is the DGD of a single section in this case. When the width is progressively increased to 30%, we see that the structure is completely smeared out above the 20% width, which was the width used in the simulations henceforth. The model simulations with unequal segments predict nonperiodic variations of the DGD, and its statistics over a 100-nm spectral width gets close to Maxwellian characteristics even for a single set of $\{h_n\}$, $\{\alpha_n\}$, and $\{\phi_n\}$. The calculated statistical distribution of the DGD values of an 8.7-km-long SMF with a PMD coefficient of 9.15 ps/ $\sqrt{\text{km}}$ is presented in Fig. 2(a). The distribution was obtained from a single DGD spectrum, in the 1480–1580 nm spectral window and a wavelength step of 0.1 nm, corresponding to a single set of $N = 50$ sections $\{h_n\}$, coupling angles $\{\alpha_n\}$, and phases $\{\phi_n\}$. It is clear that the statistics is well described by a Maxwellian fit, meaning that our approach to the stochastic characteristics of the phenomenon is correct. It should be noted that the DGD statistics of a real fiber experimentally obtained from a single limited wavelength scan is close but not exactly Maxwellian due to the limited statistics. Keeping the same lengths $\{h_n\}$ and coupling angles $\{\alpha_n\}$, but randomly varying the phase angles $\{\phi_n\}$ we obtain statistical distributions which are similar to the one presented in Fig. 2(a), their mean value differing by less than 3%, a situation that is quite similar to the measurement of a SMF in different environmental conditions. However, different sets $\{h_n\}$ and $\{\alpha_n\}$ give rise to different mean values of the DGD's around the $\langle\Delta\tau\rangle = 27$ ps mean value, given by the PMD coefficient and fiber length. Although all these sets have the same mean segment length and PMD coefficient, their PMD was found to follow a Gaussian distribution with a standard deviation which depends of the number of sections used in the simulation. This standard deviation decreases from 20% and stabilizes at 4% of the goal PMD value $\langle\Delta\tau\rangle$ as the number of sections is increased from 20 to 1000.

III. EXPERIMENTAL ANALYSIS

An 8.7-km-long DSF and a 20-sections PMD emulator, had their DGD spectra measured using the modulation phase shift (MPS) technique [12], within the mode hopping-free 2-nm

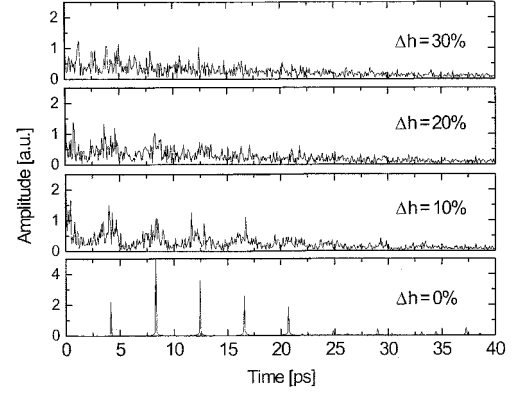


Fig. 1. Fourier transform of the DGD spectral dependency simulated with 50 unequal sections and different standard deviations Δh around the mean section length.

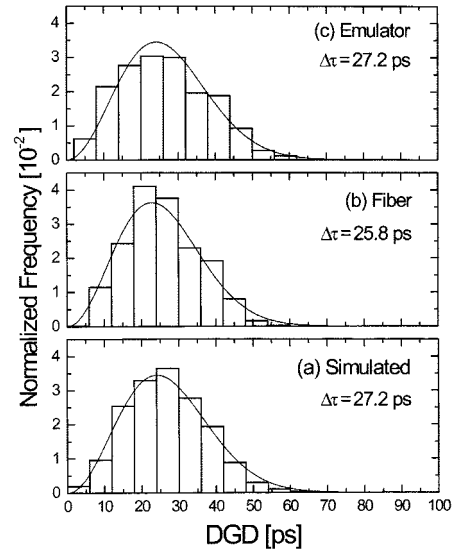


Fig. 2. (a) Statistical distribution of simulated DGD in an 8.7-km-long fiber with 9.15 ps/ $\sqrt{\text{km}}$ PMD coefficient. A single set of $N = 50$ coupling-lengths, angles and phases were used and the statistics were obtained in the 1480–1580 nm spectral window, (b) statistical distribution of measured DGD's in an 8.7-km-long DSF with the same PMD coefficient, and (c) statistical distribution of measured DGD's in a PMD emulator with the same total PMD as the fiber.

spectral range of a tunable DBR laser, directly modulated at 1 GHz. The emulator was built up with 20 pieces of HiBi fiber with random lengths with 20% spread as described in [10]. Both had the same 27 ps total PMD measured with the interferometric technique. The statistical distributions of the measured DGD's in the fiber and PMD emulator are presented in Fig. 2(b) and (c). Because of the slowly varying room temperature the principal states and DGD of the emulator or fiber are also varying with time and wavelength, so that the ensemble of DGD's varies and a statistical distribution is obtained combining time and wavelength scanning. Each histogram was obtained from 22 DGD spectral measurements, each of them in the 1545–1547-nm spectral window. The mean wavelength step was 0.1 nm providing ~ 500 independent data points. In each spectral measurement, it was observed that the shape of DGD curve has a nonzero correlation with the previous one. However, as the room temperature and environmental conditions vary, the DGD curves change, and the correlation

is lost. Hence, in order to have a random set of DGD values, we performed the DGD spectral measurements at different times and room temperatures in a two-day experiment. The histograms in Fig. 2 are well fitted by Maxwellian distributions, with mean values in good agreement with the interferometric technique measurements.

Second-order PMD is given not only by the DGD spectral dependency but also by the spectral variation of the PSP. Our measurement apparatus is unable to measure the PSP, but the results for the DGD spectra provide data on the frequency derivative of the DGD's, $\partial(\delta\tau)/\partial\omega$, which is of the order of 1/9th of the total second-order PMD [3]. Although this is only a partial approach to the problem the results are directly obtained from the experimental DGD spectra and can be easily compared. Fig. 3 displays the statistical distribution of the DGD frequency derivatives simulated with a 50-sections model with $\Delta h = 20\%$, compared with the distributions obtained from the fiber and emulator spectral DGD measurements. The three distributions were obtained with the same 0.1-nm wavelength step, which was the one used in the experimental measurements. We actually tested the resolution running the theoretical model with smaller steps and found that the actual distribution has twice the observed width indicating that the wavelength step used in the experiments is too large, leading to an underestimation of the DGD derivatives. However, no further increase in the experimental wavelength resolution could be done because of the laser chirp arising from the direct RF modulation. Nevertheless, the general agreement between the three statistical distributions presented in Fig. 3 is clear, indicating that the model simulations and the emulator provide good descriptions of the first- and second-order PMD effects in a real fiber. Of course, the PMD emulator behaves statistically, and long times are needed to cover a good statistics when the environmental conditions are stable. This drawback can be easily overcome if these conditions are forced to vary or a wider spectral window is used.

IV. CONCLUSION

We show experimentally and theoretically that with the random coupling-length model, a single set of coupling-lengths and angles accounts for both, the Maxwellian statistics and the nonperiodical DGD spectral dependence. Different environmental conditions such as the fiber temperature distribution are simulated by a random phase factor at the coupling points. Different phase sets present different DGD spectra with the same Maxwellian statistics just as independent measurements of the same fiber. Depending on the number of sections used, different sets of coupling-lengths and angles with the same mean coupling-length will actually describe different PMD values converging within 4% to the goal PMD, given by the PMD coefficient and total fiber length.

Finally, we show that a device made by a concatenation of HiBi fibers segments with random lengths and coupling angles can emulate the DGD wavelength dependence of SMF's and its statistics as well. We found good agreement between our model simulations and the experimental results obtained either with the emulator device or with an 8.7-km DSF with the same PMD. We conclude that both, the random coupling-lengths mathematical

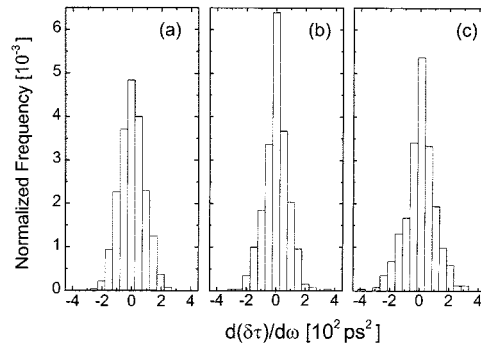


Fig. 3. Statistical distribution of DGD frequency derivatives simulated with a 50-section model as in Fig. 2: (a) compared to the measured distributions in the 8.7 km dispersion-shifted fiber (b) and the PMD emulator (c).

model and the emulator device provide good descriptions either for the first-order PMD statistics or second-order PMD and are powerful tools for the simulation of signal distortion due to these phenomena.

ACKNOWLEDGMENT

The authors would like to acknowledge F. S. Vianna and L. C. B. Linares for helping with the experiments.

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